

Integrative analysis for sc-/sn-RNA-Seq Analysis in 10X Genomics data

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```
## The following objects are masked _by_ .GlobalEnv:
##
##   PCA_Feature_List_value, Regress_Cell_Cycle_value, assay,
##   genome_name, integration_method, nfeatures_value, root_dir
```

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DNB Bioinformatics Core Pipeline: Standard sc-/sn-RNA-Seq Analysis in 10X Genomics data

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1 Information about this notebook

Biological heterogeneity in single-cell RNA-seq data is often confounded by technical factors including sequencing depth. The number of molecules detected in each cell can vary significantly between cells, even within the same celltype. Interpretation of scRNA-seq data requires effective pre-processing and normalization to remove this technical variability (for more information, see also scRNA-seq Dataset Integration²).

Hence, integration of single-cell sequencing datasets, for example across experimental batches, donors, or conditions, is often an important step in scRNA-seq workflows. Integrative analysis can help to match shared cell types and states across datasets, which can boost statistical power, and most importantly, facilitate accurate comparative analysis across datasets. Here, we include the following integration methods for integrative analysis:

- (1) Seurat ‘anchor-based’³ integration workflow by Choudhary & Satija, 2022⁴,
- (2) Harmony⁵ integration workflow by Korsunsky et al., 2019⁶, and
- (3) Liger⁷ integration workflow by Welch et al., 2019⁸.

Different integration algorithms, such as Seurat, Harmony, and Liger, often use different numbers of dimensions. The optimal number of dimensions depends on the specific algorithm and its associated parameters. While these algorithms generally consider similar variables, the mathematical approaches they use to process and transform those variables differ significantly, which means the same number of dimensions (or “resolutions”) doesn’t work equally well across all algorithms.

Even though they all refer to “resolution,” the effect of resolution can vary from one algorithm to another, and the maximum number of dimensions they can effectively use may differ. Based on our experience, we recommend using 30 dimensions for Seurat, 30 for Harmony, and 20 for Liger. These settings tend to produce the most consistent integration results without unnecessarily increasing computational load.

2 Set up

```
attach(params)
```

```
## The following objects are masked _by_ .GlobalEnv:  
##  
##   PCA_Feature_List_value, Regress_Cell_Cycle_value, assay,  
##   genome_name, integration_method, nfeatures_value, root_dir
```

```
suppressPackageStartupMessages({  
  library(future)  
  library(tidyverse)  
  library(patchwork)  
  library(Seurat)  
  library(SeuratObject)  
  library(harmony)
```

²<https://www.singlecellcourse.org/scrna-seq-dataset-integration.html#seurat-v3-3-vs-5-10k-pbmc>

³https://satijalab.org/seurat/articles/seurat5_integration

⁴<https://doi.org/10.1186/s13059-021-02584-9>

⁵<https://portals.broadinstitute.org/harmony/>

⁶<https://www.nature.com/articles/s41592-019-0619-0>

⁷https://welch-lab.github.io/liger/articles/liger_with_seurat.html

⁸<https://doi.org/10.1016/j.cell.2019.05.006>

```

library(rliger)
library(RcppPlanc)
library(SeuratWrappers)
library(scooter)
library(reshape2)
library(RColorBrewer)
library(knitr)

# Evaluate Seurat R expressions asynchronously when possible using future package
options(future.globals.maxSize = future_globals_value)
plan(multisession, workers = parallelly::availableCores())}

```

3 Directories and paths to file Inputs/Outputs

```

#analysis_dir <- file.path(root_dir, "analyses", "integrative-analysis")
#data_dir <- file.path(root_dir, "analyses", "upstream-analysis", "results", "04_Filter_obj")
#upstream_analysis_dir <- file.path(root_dir, "analyses", "upstream-analysis")

# Input files
#data_file <- file.path(data_dir, "seurat_obj_merged_filtered.rds")

# Create results_dir
#results_dir <- file.path(analysis_dir, "results")
#if (!dir.exists(results_dir)) {
#  dir.create(results_dir)}

# Create plots directory
#report_dir <- file.path(analysis_dir, "plots")
#if (!dir.exists(report_dir)) {
#  dir.create(report_dir)}

#plots_dir <- file.path(report_dir, glue::glue("plots_{integration_method}"))
#if (!dir.exists(plots_dir)) {
#  dir.create(plots_dir)}

source(paste0(root_dir, "/figures/scripts/theme_plot.R"))
#source(paste0(analysis_dir, "/util/function-samples-integrate.R"))
#source(paste0(analysis_dir, "/util/custom-seurat-functions.R"))
#source(paste0(upstream_analysis_dir, "/util/function-process-Seurat.R"))

source(paste0(module_dir, "/util/function-samples-integrate.R"))
source(paste0(module_dir, "/util/custom-seurat-functions.R"))
source(paste0(module_dir, "/util/function-process-Seurat.R"))

```

4 Read seurat object

First, we will use the `seurat_obj_merged_filtered.rds` object as generated from the pipeline in the `upstream-analysis` module. This object contains all samples of the project merged.


```
merged_obj <- readRDS(data_file)
```

5 Seurat ‘anchor-based’ integration workflow

Seurat integration creates a unified object that contains both original data (‘RNA’ assay) as well as integrated data (‘integrated’ assay).

5.1 Normal-sized datasets

SCT method (or `sctransform`) is a modeling framework for the normalization and variance stabilization of molecular count data from scRNA-seq experiments. This procedure omits the need for heuristic steps including pseudocount addition or log-transformation and improves common downstream analytical tasks such as variable gene selection, dimensional reduction, and differential expression. More specifically, the `sctransform` approach utilizes the Pearson residuals from negative binomial regression as input to standard dimensional reduction techniques.

5.2 Large-sized datasets

For very large datasets, the standard integration workflow can sometimes be prohibitively computationally expensive. In addition, Seurat and a lot of integration algorithms are designed to integrate data up to a maximum amount. Here, we use the `big_data` parameter as an option for integration of extremely large datasets. For example, 1.3 millions cells are too much and so this option exist for those datasets. The problem comes down to infinite matrix sizes and memory issues, no computer to data can compute this many variables so new algorithms exist to get around that through various methods. For more information, see Seurat v4 large data integration⁹ and Seurat v5 large data integration¹⁰.

A `reference_list`¹¹ can be used when conducting Seurat integration which help with speeding up integration.

Moreover, we improve efficiency and runtimes by employing (1) Reciprocal PCA (RPCA) and (2) Reference-based integration. The main efficiency improvements are gained in `FindIntegrationAnchors()`. First, we use reciprocal PCA (RPCA), to identify an effective space in which to find anchors. When determining anchors between any two datasets using reciprocal PCA, we project each dataset into the others PCA space and constrain the anchors by the same mutual neighborhood requirement. All downstream integration steps remain the same and we are able to ‘correct’ (or harmonize) the datasets.

Additionally, we use reference-based integration. In the standard workflow, we identify anchors between all pairs of datasets. While this gives datasets equal weight in downstream integration, it can also become computationally intensive. For example when integrating 10 different datasets, we perform 45 different pairwise comparisons. As an alternative, we introduce here the possibility of specifying one or more of the datasets as the ‘reference’ for integrated analysis, with the remainder designated as ‘query’ datasets. In this workflow, we do not identify anchors between pairs of query datasets, reducing the number of comparisons. For example, when integrating 10 datasets with one specified as a reference, we perform only 9 comparisons. Reference-based integration can be applied to either log-normalized or SCTransform-normalized datasets.

Note that the anchor-based RPCA integration represents a faster and more conservative (less correction) method for integration.

⁹https://satijalab.org/seurat/archive/v4.3/integration_large_datasets

¹⁰https://satijalab.org/seurat/articles/seurat5_sketch_analysis

¹¹<https://satijalab.org/seurat/reference/findintegrationanchors#examples>

```

# We will use the Seurat 'anchor-based' workflow, if defined in `params`.
if (use_seurat_integration == "YES"){
  print_message <- "we use the Seurat 'anchor-based' workflow"

  # Make seurat_list
  seurat_obj_list <- list()
  seurat_obj_list <- SplitObject(merged_obj, split.by = "orig.ident")

  # Run function
  seurat_obj <- seurat_integration(seurat_obj_list = seurat_obj_list,
                                   nfeatures_value = nfeatures_value,
                                   num_dim_seurat = num_dim_seurat,
                                   num_dim_seurat_integration = num_dim_seurat_integration,
                                   big_data = big_data_value,
                                   Genome = genome_name,
                                   Regress_Cell_Cycle = Regress_Cell_Cycle_value,
                                   reference_list = reference_list_value,
                                   PCA_Feature_List = PCA_Feature_List_value)
} else {
  print_message <- "we will skip usage of the Seurat 'anchor-based' workflow"
}

```

Here, we will skip usage of the Seurat 'anchor-based' workflow.

6 Harmony integration workflow

Harmony is a general-purpose R package with an efficient algorithm for integrating multiple data sets. It is especially useful for large single-cell datasets such as single-cell RNA-seq. Harmony algorithm projects cells into a shared embedding in which cells group by cell type rather than dataset-specific conditions. Harmony simultaneously accounts for multiple experimental and biological factors. We will run Harmony within our Seurat workflow.

```

# We will use the Harmony integration workflow, if defined in `params`.
if (use_harmony_integration == "YES"){
  print_message <- "we use the Harmony integration workflow"
  seurat_obj <- harmony_integration(seurat_obj = merged_obj,
                                    variables_to_integrate = variable_value,
                                    num_dim = num_dim_harmony)

  # Plot
  name <- paste0(plots_dir, "/", glue::glue("plot_{integration_method}.png"))
  p1 <- DimPlot(object = seurat_obj, reduction = glue::glue("{integration_method}"), pt.size = 10)
  p2 <- VlnPlot(object = seurat_obj, features = "harmony_1", group.by = "orig.ident", pt.size = 10)
  print(patchwork::wrap_plots(list(p1, p2), nrow = 1))
  ggsave(file = name, width = 12, height = 5, device = "png")
} else {
  print_message <- "we will skip usage of the Harmony integration workflow"
}

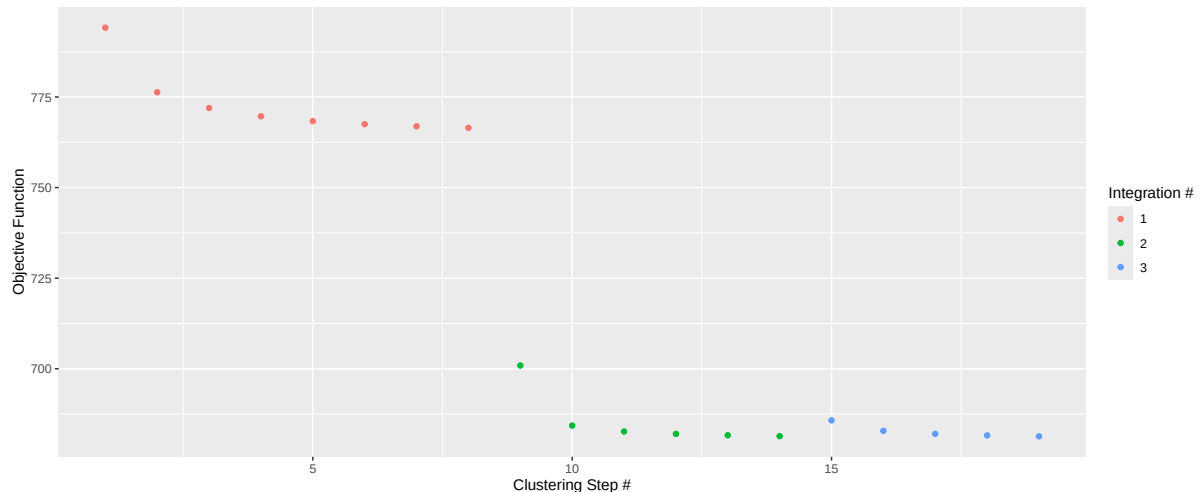
```

```
## Performing Harmony Integration
```

```
## Transposing data matrix
```

```
## Initializing state using k-means centroids initialization
```

```
## Harmony 1/10
## Harmony 2/10
## Harmony 3/10
## Harmony converged after 3 iterations
```



```
## Performing Uniform Manifold Approximation and Projection (UMAP) dimensional reduction te
## 18:16:02 UMAP embedding parameters a = 0.9922 b = 1.112
## 18:16:02 Read 45672 rows and found 30 numeric columns
## 18:16:02 Using Annoy for neighbor search, n_neighbors = 30
## 18:16:02 Building Annoy index with metric = cosine, n_trees = 50
## 0% 10 20 30 40 50 60 70 80 90 100%
## [----|----|----|----|----|----|----|----|----|----|
## *****|
## 18:16:07 Writing NN index file to temp file /lsf_tmp/264625707.tmpdir/RtmpbhtMvE/file2f9
## 18:16:07 Searching Annoy index using 16 threads, search_k = 3000
## 18:16:08 Annoy recall = 100%
## 18:16:09 Commencing smooth kNN distance calibration using 16 threads with target n_neigh
## 18:16:10 Initializing from normalized Laplacian + noise (using irlba)
## 18:16:11 Commencing optimization for 200 epochs, with 1949692 positive edges
## 18:16:30 Optimization finished

## Generate and add metadata

##
##
## ===== saving metadata =====
##
##
##
##
## ===== saving metadata =====

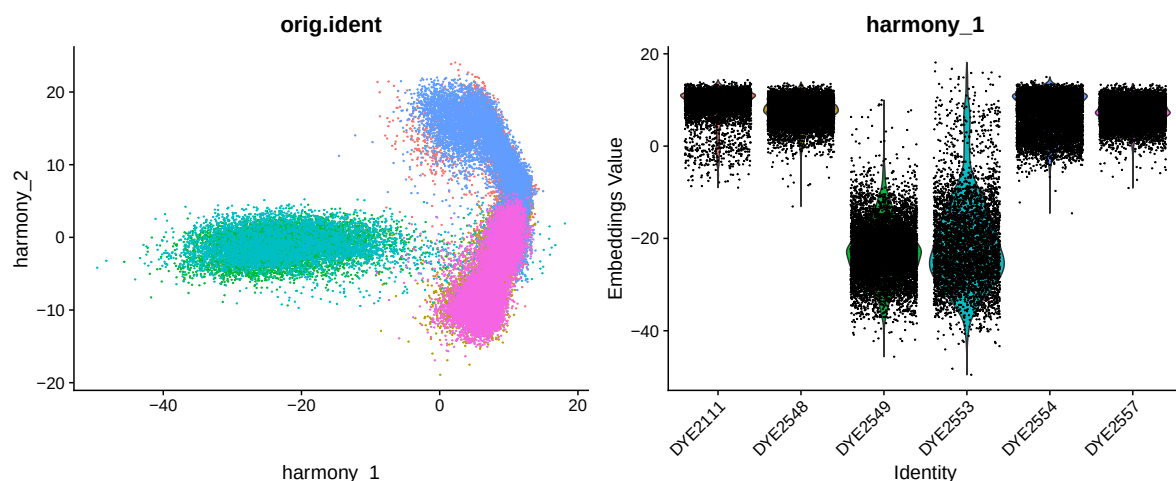
## Computing the k.param nearest neighbors for a given dataset and clusters
```

```
## Computing nearest neighbor graph
##Computing SNN

##Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck
##
##Number of nodes: 45672
##Number of edges: 1512225
##
##Running Louvain algorithm...
##Maximum modularity in 10 random starts: 0.8793
##Number of communities: 21
##Elapsed time: 12 seconds

##1 singletons identified. 20 final clusters.

##Returned Seurat object
```



Here, we use the Harmony integration workflow.

7 Liger integration workflow

LIGER is an algorithm that enables the identification of shared cell types across individuals, species, and multiple modalities (gene expression, epigenetic or spatial data), as well as dataset-specific features, offering a unified analysis of heterogeneous single-cell datasets.

More specifically, LIGER can be used to integrate datasets: (1) from multiple experiments and biological contexts (differences in technical variation, e.g., number of cells sampled, or in the depth of sequencing allocated to each cell), (2) across space (in situ and single-cell expression data to spatially locate fine subtypes of cells) and (3) various modalities (single-cell RNA-seq and DNA methylation or epigenomes). Across modalities, datasets may vary widely in dynamic range (gene expression versus chromatin accessibility), direction of relationship (RNA-seq versus DNA methylation), or in the number of genes measured (targeted quantification versus unbiased approaches).

```
# We will use the Liger workflow, if defined in `params`.
if (use_liger_integration == "YES"){
  print_message <- "we use the Liger workflow"
  seurat_obj <- liger_integration(seurat_obj = merged_obj,
                                variables_to_integrate = variable_value,
```

```

n_neighbors = n_neighbors_value)

# Plot
name <- paste0(plots_dir, "/", glue::glue("plot_{integration_method}.png"))
DimPlot(seurat_obj, reduction = glue::glue("{integration_method}"), group.by = c("orig.ident"),
ggsave(file = name, width = 6, height = 5, device = "png")

} else {
  print_message <- "we will skip usage of the Liger workflow"}

```

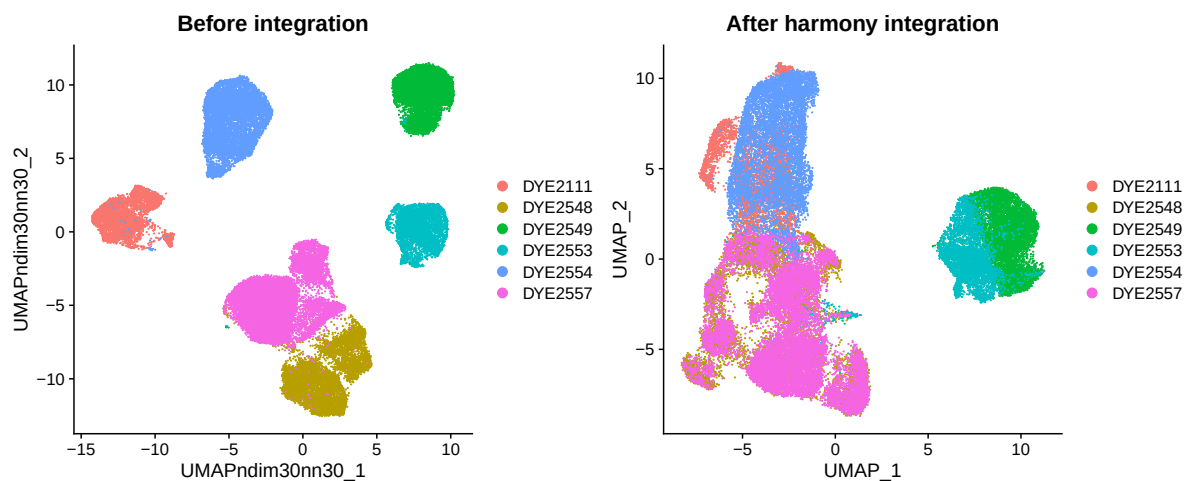
Here, we will skip usage of the Liger workflow.

8 Before vs After integration plot

```

# Plot
name <- paste0(plots_dir, "/", glue::glue("plot_unintegrated_vs_{integration_method}.png"))
p1 <- DimPlot(merged_obj, group.by = c("orig.ident")) + ggtitle("Before integration")
p2 <- DimPlot(seurat_obj, reduction = "umap", group.by = c("orig.ident")) + ggtitle(glue::glue("{integration_method}"))
patchwork::wrap_plots(list(p1, p2), nrow = 1)

```



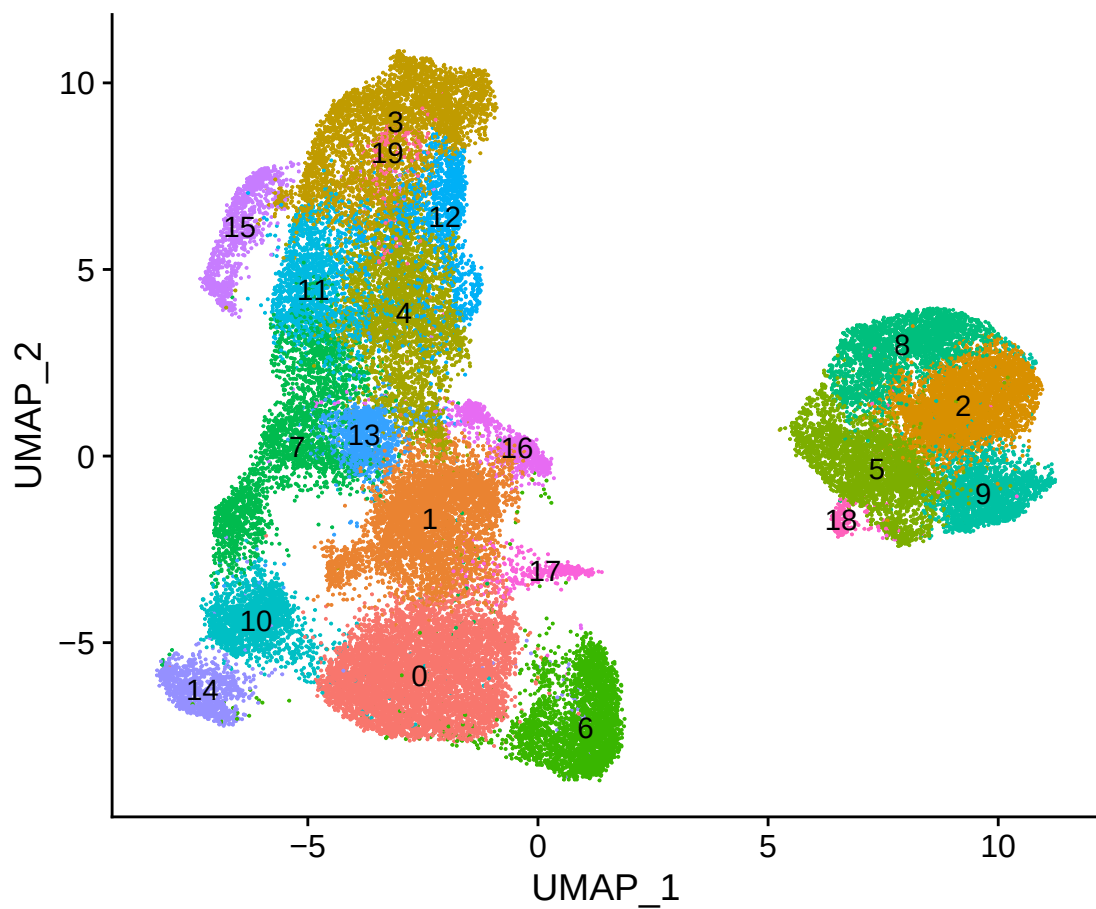
```
ggsave(file = name, width = 12, height = 5, device = "png")
```

```
seurat_obj <- SetIdent(seurat_obj, value = "seurat_clusters")
```

```

# Plot
name <- paste0(plots_dir, "/", glue::glue("plot_clusters_{integration_method}.png"))
DimPlot(seurat_obj, reduction = "umap", label = TRUE) + NoLegend()

```

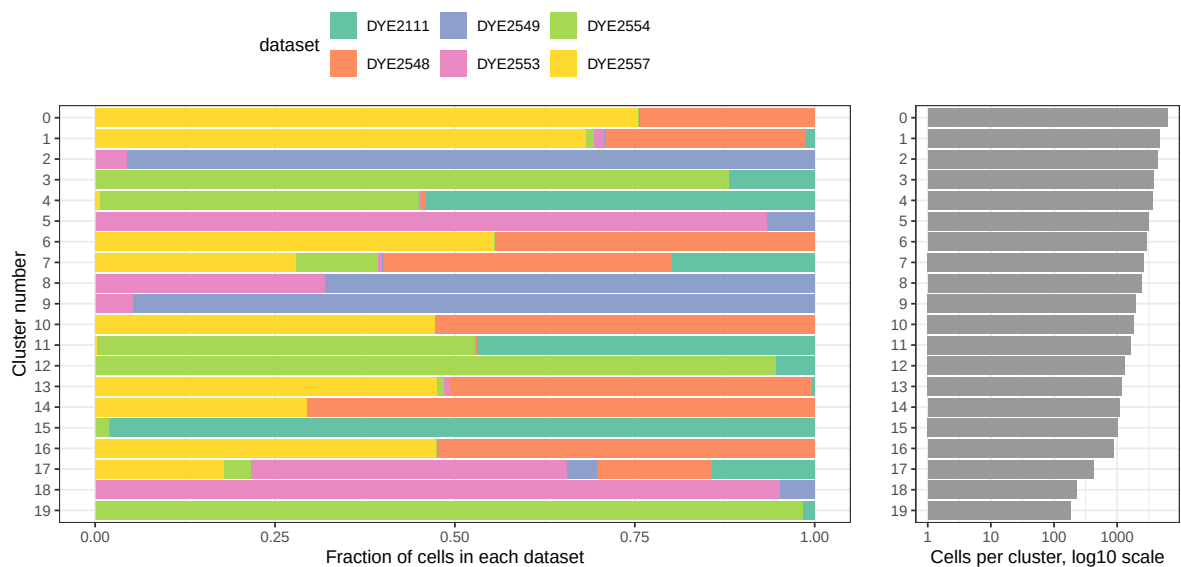


```
ggsave(file = name, width = 6, height = 5, device = "png")
```

```
# Plot
```

```
name <- paste0(plots_dir, "/", glue::glue("plot_fraction_of_cells_{integration_method}.png"),  
plot_integrated_clusters(seurat_obj)
```

```
## Using cluster as id variables
```



```
ggsave(file = name, width = 10, height = 6, device = "png")
```

9 Save output files

```
# Identify columns with a '.1' suffix
cols_to_remove <- grep("\\.1$", colnames(seurat_obj@meta.data), value = TRUE)

# Exclude columns that match the specific patterns (e.g., {assay}_snn_res.0.1, {assay}_snn_res.1.0)
cols_to_remove <- cols_to_remove[!grepl(glue::glue("^.{assay}_snn_res\\.0\\.1$"), cols_to_remove) &&
!grepl(glue::glue("^.{assay}_snn_res\\.1\\.0$"), cols_to_remove) &&
!grepl(glue::glue("^.{assay}_snn_res\\.1\\.10$"), cols_to_remove)]

# Remove the columns
seurat_obj@meta.data <- seurat_obj@meta.data[, !colnames(seurat_obj@meta.data) %in% cols_to_remove]
head(seurat_obj@meta.data)
```

```
##               orig.ident nCount_RNA nFeature_RNA percent.mito
## DYE2111:AAACCCAAGAGGTATT   DYE2111      34571         6588      2.895
## DYE2111:AAACCCAAGGAAGCTCG   DYE2111      49746         7951      3.052
## DYE2111:AAACCCAAGTCATAGA     DYE2111      23386         5694      2.972
## DYE2111:AAACCCATCAAGCCTA     DYE2111      26406         5455      2.458
## DYE2111:AAACCCATCATGCCGG     DYE2111      25219         5011      2.661
## DYE2111:AAACCCATCGGCTTCT     DYE2111      50953         8142      2.237
##               log10GenesPerUMI      ID      SAMPLE
## DYE2111:AAACCCAAGAGGTATT      0.8413739 DYE2111 3149501_DYE2111
## DYE2111:AAACCCAAGGAAGCTCG      0.8304498 DYE2111 3149501_DYE2111
## DYE2111:AAACCCAAGTCATAGA      0.8595686 DYE2111 3149501_DYE2111
## DYE2111:AAACCCATCAAGCCTA      0.8451031 DYE2111 3149501_DYE2111
## DYE2111:AAACCCATCATGCCGG      0.8405618 DYE2111 3149501_DYE2111
## DYE2111:AAACCCATCGGCTTCT      0.8308031 DYE2111 3149501_DYE2111
##
## DYE2111:AAACCCAAGAGGTATT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/
1/3149501/
## DYE2111:AAACCCAAGGAAGCTCG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/
1/3149501/
## DYE2111:AAACCCAAGTCATAGA /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/
1/3149501/
## DYE2111:AAACCCATCAAGCCTA /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/
1/3149501/
## DYE2111:AAACCCATCATGCCGG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/
1/3149501/
## DYE2111:AAACCCATCGGCTTCT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/
1/3149501/
##               SJUID      SJID      cancer_type
## DYE2111:AAACCCAAGAGGTATT SJHBBBQCTFK SJEWS056156 Ewing_sarcoma
## DYE2111:AAACCCAAGGAAGCTCG SJHBBBQCTFK SJEWS056156 Ewing_sarcoma
## DYE2111:AAACCCAAGTCATAGA SJHBBBQCTFK SJEWS056156 Ewing_sarcoma
## DYE2111:AAACCCATCAAGCCTA SJHBBBQCTFK SJEWS056156 Ewing_sarcoma
## DYE2111:AAACCCATCATGCCGG SJHBBBQCTFK SJEWS056156 Ewing_sarcoma
## DYE2111:AAACCCATCGGCTTCT SJHBBBQCTFK SJEWS056156 Ewing_sarcoma
```

##	Subject_Disease_Code	Subject_Disease_Subtype_Code		
## DYE2111:AAACCCAAGAGGTATT	EWS	EWS		
## DYE2111:AAACCCAAGGAACTCG	EWS	EWS		
## DYE2111:AAACCCAAGTCATAGA	EWS	EWS		
## DYE2111:AAACCCATCAAGCCTA	EWS	EWS		
## DYE2111:AAACCCATCATGCCGG	EWS	EWS		
## DYE2111:AAACCCATCGGCTTCT	EWS	EWS		
##	tissue_type	assay_technology_updated	assay_technology	
## DYE2111:AAACCCAAGAGGTATT	CCLF_cells	10x_scrNaseq	10x_scrNaseq	
## DYE2111:AAACCCAAGGAACTCG	CCLF_cells	10x_scrNaseq	10x_scrNaseq	
## DYE2111:AAACCCAAGTCATAGA	CCLF_cells	10x_scrNaseq	10x_scrNaseq	
## DYE2111:AAACCCATCAAGCCTA	CCLF_cells	10x_scrNaseq	10x_scrNaseq	
## DYE2111:AAACCCATCATGCCGG	CCLF_cells	10x_scrNaseq	10x_scrNaseq	
## DYE2111:AAACCCATCGGCTTCT	CCLF_cells	10x_scrNaseq	10x_scrNaseq	
##	historic_data	RPMI	genome_reference_updated_code	
## DYE2111:AAACCCAAGAGGTATT	no	no	human-genome	
## DYE2111:AAACCCAAGGAACTCG	no	no	human-genome	
## DYE2111:AAACCCAAGTCATAGA	no	no	human-genome	
## DYE2111:AAACCCATCAAGCCTA	no	no	human-genome	
## DYE2111:AAACCCATCATGCCGG	no	no	human-genome	
## DYE2111:AAACCCATCGGCTTCT	no	no	human-genome	
##	genome_reference_updated	genome_reference		
## DYE2111:AAACCCAAGAGGTATT	human-genome	dual-genome		
## DYE2111:AAACCCAAGGAACTCG	human-genome	dual-genome		
## DYE2111:AAACCCAAGTCATAGA	human-genome	dual-genome		
## DYE2111:AAACCCATCAAGCCTA	human-genome	dual-genome		
## DYE2111:AAACCCATCATGCCGG	human-genome	dual-genome		
## DYE2111:AAACCCATCGGCTTCT	human-genome	dual-genome		
##				
## DYE2111:AAACCCAAGAGGTATT	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
## DYE2111:AAACCCAAGGAACTCG	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
## DYE2111:AAACCCAAGTCATAGA	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
## DYE2111:AAACCCATCAAGCCTA	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
## DYE2111:AAACCCATCATGCCGG	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
## DYE2111:AAACCCATCGGCTTCT	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
##	DYExxxx	DYE_Melody_unsure	DYExxxx_corrected	
## DYE2111:AAACCCAAGAGGTATT	DYE2111	<NA>	DYE2111	
## DYE2111:AAACCCAAGGAACTCG	DYE2111	<NA>	DYE2111	
## DYE2111:AAACCCAAGTCATAGA	DYE2111	<NA>	DYE2111	
## DYE2111:AAACCCATCAAGCCTA	DYE2111	<NA>	DYE2111	
## DYE2111:AAACCCATCATGCCGG	DYE2111	<NA>	DYE2111	
## DYE2111:AAACCCATCGGCTTCT	DYE2111	<NA>	DYE2111	
##	SRM_updated	SRM	SRM_Sample_ID_updated	SRM_Sample_ID
## DYE2111:AAACCCAAGAGGTATT	829202	829202	3149501	3149501
## DYE2111:AAACCCAAGGAACTCG	829202	829202	3149501	3149501

## DYE2111:AAACCCAAGTCATAGA	829202	829202	3149501	3149501
## DYE2111:AAACCCATCAAGCCTA	829202	829202	3149501	3149501
## DYE2111:AAACCCATCATGCCGG	829202	829202	3149501	3149501
## DYE2111:AAACCCATCGGCTTCT	829202	829202	3149501	3149501
##	Experimental_Sample	sample_3D		
## DYE2111:AAACCCAAGAGGTATT	SJ1044	No		
## DYE2111:AAACCCAAGGAACTCG	SJ1044	No		
## DYE2111:AAACCCAAGTCATAGA	SJ1044	No		
## DYE2111:AAACCCATCAAGCCTA	SJ1044	No		
## DYE2111:AAACCCATCATGCCGG	SJ1044	No		
## DYE2111:AAACCCATCGGCTTCT	SJ1044	No		
##	sample_2D.NOT.NEEDED.2025.04.10	Tissue_Bank_ID		
## DYE2111:AAACCCAAGAGGTATT	No	TB-19-21394		
## DYE2111:AAACCCAAGGAACTCG	No	TB-19-21394		
## DYE2111:AAACCCAAGTCATAGA	No	TB-19-21394		
## DYE2111:AAACCCATCAAGCCTA	No	TB-19-21394		
## DYE2111:AAACCCATCATGCCGG	No	TB-19-21394		
## DYE2111:AAACCCATCGGCTTCT	No	TB-19-21394		
##	Xenograft_CompBio_ID	CSTN_MAST_ID		Date
## DYE2111:AAACCCAAGAGGTATT	SJEWS056156_X1	MAST601E	2024-06-14T00:00:00Z	
## DYE2111:AAACCCAAGGAACTCG	SJEWS056156_X1	MAST601E	2024-06-14T00:00:00Z	
## DYE2111:AAACCCAAGTCATAGA	SJEWS056156_X1	MAST601E	2024-06-14T00:00:00Z	
## DYE2111:AAACCCATCAAGCCTA	SJEWS056156_X1	MAST601E	2024-06-14T00:00:00Z	
## DYE2111:AAACCCATCATGCCGG	SJEWS056156_X1	MAST601E	2024-06-14T00:00:00Z	
## DYE2111:AAACCCATCGGCTTCT	SJEWS056156_X1	MAST601E	2024-06-14T00:00:00Z	
##	SRM_Submission_Information			
## DYE2111:AAACCCAAGAGGTATT	<NA>			
## DYE2111:AAACCCAAGGAACTCG	<NA>			
## DYE2111:AAACCCAAGTCATAGA	<NA>			
## DYE2111:AAACCCATCAAGCCTA	<NA>			
## DYE2111:AAACCCATCATGCCGG	<NA>			
## DYE2111:AAACCCATCGGCTTCT	<NA>			
##	Compbioname_for_publications	Comp_Bio_Sample_Name		
## DYE2111:AAACCCAAGAGGTATT	SJEWS056156_C3	SJEWS056156_C3-		
SJ1044T_2				
## DYE2111:AAACCCAAGGAACTCG	SJEWS056156_C3	SJEWS056156_C3-		
SJ1044T_2				
## DYE2111:AAACCCAAGTCATAGA	SJEWS056156_C3	SJEWS056156_C3-		
SJ1044T_2				
## DYE2111:AAACCCATCAAGCCTA	SJEWS056156_C3	SJEWS056156_C3-		
SJ1044T_2				
## DYE2111:AAACCCATCATGCCGG	SJEWS056156_C3	SJEWS056156_C3-		
SJ1044T_2				
## DYE2111:AAACCCATCGGCTTCT	SJEWS056156_C3	SJEWS056156_C3-		
SJ1044T_2				
##	Comp_Bio_Subject_ID	Preferred_Sample_ID		
## DYE2111:AAACCCAAGAGGTATT	SJ056156	SJ1044T_2		
## DYE2111:AAACCCAAGGAACTCG	SJ056156	SJ1044T_2		
## DYE2111:AAACCCAAGTCATAGA	SJ056156	SJ1044T_2		
## DYE2111:AAACCCATCAAGCCTA	SJ056156	SJ1044T_2		
## DYE2111:AAACCCATCATGCCGG	SJ056156	SJ1044T_2		

##	DYE2111:AAACCCATCGGCTTCT	SJ056156	SJ1044T_2		
##		Sample_Type_Code	Prior_Comp_Bio_ID	PATH	NOTES
##	DYE2111:AAACCCAAGAGGTATT	C	SJEWS056156	<NA>	NA
##	DYE2111:AAACCCAAGGAACTCG	C	SJEWS056156	<NA>	NA
##	DYE2111:AAACCCAAGTCATAGA	C	SJEWS056156	<NA>	NA
##	DYE2111:AAACCCATCAAGCCTA	C	SJEWS056156	<NA>	NA
##	DYE2111:AAACCCATCATGCCGG	C	SJEWS056156	<NA>	NA
##	DYE2111:AAACCCATCGGCTTCT	C	SJEWS056156	<NA>	NA
##		match_SJID_RPMI	ID_with_sample_swap		
##	DYE2111:AAACCCAAGAGGTATT	SJEWS056156_no	DYE2111		
##	DYE2111:AAACCCAAGGAACTCG	SJEWS056156_no	DYE2111		
##	DYE2111:AAACCCAAGTCATAGA	SJEWS056156_no	DYE2111		
##	DYE2111:AAACCCATCAAGCCTA	SJEWS056156_no	DYE2111		
##	DYE2111:AAACCCATCATGCCGG	SJEWS056156_no	DYE2111		
##	DYE2111:AAACCCATCGGCTTCT	SJEWS056156_no	DYE2111		
##		SAMPLE_with_sample_swap			
##	DYE2111:AAACCCAAGAGGTATT	3149501_DYE2111			
##	DYE2111:AAACCCAAGGAACTCG	3149501_DYE2111			
##	DYE2111:AAACCCAAGTCATAGA	3149501_DYE2111			
##	DYE2111:AAACCCATCAAGCCTA	3149501_DYE2111			
##	DYE2111:AAACCCATCATGCCGG	3149501_DYE2111			
##	DYE2111:AAACCCATCGGCTTCT	3149501_DYE2111			
##	DYE2111:AAACCCAAGAGGTATT	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
##	DYE2111:AAACCCAAGGAACTCG	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
##	DYE2111:AAACCCAAGTCATAGA	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
##	DYE2111:AAACCCATCAAGCCTA	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
##	DYE2111:AAACCCATCATGCCGG	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
##	DYE2111:AAACCCATCGGCTTCT	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/			
##		S.Score	G2M.Score	Phase	UMAP_1 UMAP_2
##	DYE2111:AAACCCAAGAGGTATT	0.055396504	0.5217267	G2M	0.7912828 6.248889
##	DYE2111:AAACCCAAGGAACTCG	0.029724122	0.4497271	G2M	1.9758112 7.561102
##	DYE2111:AAACCCAAGTCATAGA	0.003884657	0.1267804	G2M	1.9600578 4.757619
##	DYE2111:AAACCCATCAAGCCTA	0.194223495	-0.2824893	S	4.2224048 5.308984
##	DYE2111:AAACCCATCATGCCGG	0.005253221	0.4780977	G2M	1.5425332 5.045206
##	DYE2111:AAACCCATCGGCTTCT	-0.060120228	0.3925691	G2M	2.1266411 8.001167
##		PC_1	PC_2	PC_3	PC_4 PC_5
##	DYE2111:AAACCCAAGAGGTATT	10.253152	-22.68979	-0.6323919	-0.9587824 8.769602
##	DYE2111:AAACCCAAGGAACTCG	8.485662	-24.39137	0.8013676	-1.4042601 10.082944
##	DYE2111:AAACCCAAGTCATAGA	10.951001	-20.07229	-1.7588574	0.2910809 7.229904
##	DYE2111:AAACCCATCAAGCCTA	11.597054	-18.91220	-0.3046808	1.4613042 4.900138
##	DYE2111:AAACCCATCATGCCGG	13.183820	-23.07905	-3.7541129	-0.3845318 10.246315
##	DYE2111:AAACCCATCGGCTTCT	8.375762	-21.93434	2.5777548	-1.4043628 7.400935
##		PC_6	PC_7	PC_8	PC_9
##	DYE2111:AAACCCAAGAGGTATT	-2.7712517	-9.366240	-1.4974979	2.72132666

## DYE2111:AAACCCAAGGAACTCG	-0.3636017	-12.209703	-4.7558793	1.74341078
## DYE2111:AAACCCAAGTCATAGA	3.1424492	-7.601283	-0.1429734	2.26775331
## DYE2111:AAACCCATCAAGCCTA	8.6664426	-8.547095	-3.4713420	0.05574464
## DYE2111:AAACCCATCATGCCGG	0.3128649	-7.595591	1.7437701	1.86239495
## DYE2111:AAACCCATCGGCTTCT	-0.9476664	-11.345271	-3.7922004	0.40670382
##	PC_10	PC_11	PC_12	PC_13
## DYE2111:AAACCCAAGAGGTATT	1.23112598	1.582408	-0.46662568	-2.789662
## DYE2111:AAACCCAAGGAACTCG	3.70481977	3.357483	-0.22436947	-3.751007
## DYE2111:AAACCCAAGTCATAGA	0.08998602	2.052672	-0.52955431	-3.742995
## DYE2111:AAACCCATCAAGCCTA	-3.97244738	3.535814	0.08966243	-1.188321
## DYE2111:AAACCCATCATGCCGG	1.07962243	-1.074452	-0.84847652	-2.609866
## DYE2111:AAACCCATCGGCTTCT	3.38565479	3.496356	-1.30506804	-3.548699
##	PC_14	PC_15	PC_16	PC_17
## DYE2111:AAACCCAAGAGGTATT	0.126148021	-0.84918671	1.736273165	-1.7131367
## DYE2111:AAACCCAAGGAACTCG	0.059867572	0.38250345	-0.004923742	-0.6104021
## DYE2111:AAACCCAAGTCATAGA	0.324806608	0.97984809	0.379255062	-0.4690637
## DYE2111:AAACCCATCAAGCCTA	-0.667426810	-0.50624282	0.038240148	-0.8174554
## DYE2111:AAACCCATCATGCCGG	-0.006686692	-0.01481017	-0.599154869	-0.4504755
## DYE2111:AAACCCATCGGCTTCT	0.828032768	0.48924571	1.251134631	-0.4071848
##	PC_18	PC_19	PC_20	PC_21
## DYE2111:AAACCCAAGAGGTATT	0.3221185	-0.98278867	1.7082152	-1.4219440
## DYE2111:AAACCCAAGGAACTCG	-1.7462241	0.70255332	1.1868776	-1.8061401
## DYE2111:AAACCCAAGTCATAGA	0.1839820	0.05369649	1.1130911	0.2894633
## DYE2111:AAACCCATCAAGCCTA	0.8179425	0.31679649	0.5496390	-0.9057892
## DYE2111:AAACCCATCATGCCGG	-1.2985162	-1.58671698	0.3389435	1.0787569
## DYE2111:AAACCCATCGGCTTCT	-1.4768153	-0.73618128	0.3753112	-0.6263584
##	PC_22	PC_23	PC_24	PC_25
## DYE2111:AAACCCAAGAGGTATT	0.04758052	-0.75999468	-0.6758716	-1.03014705
## DYE2111:AAACCCAAGGAACTCG	-2.06483099	-0.04886289	-1.6141406	-0.34460262
## DYE2111:AAACCCAAGTCATAGA	-0.92594862	-0.67978153	-0.1869200	0.72142182
## DYE2111:AAACCCATCAAGCCTA	-0.18961110	0.69286040	-1.2093640	0.46370255
## DYE2111:AAACCCATCATGCCGG	0.07483081	-0.72005199	0.2030740	-0.93556990
## DYE2111:AAACCCATCGGCTTCT	0.05809369	0.46735142	-0.2989385	-0.08005591
##	PC_26	PC_27	PC_28	PC_29
## DYE2111:AAACCCAAGAGGTATT	0.68494819	-1.21100072	-0.6489976	-1.04018690
## DYE2111:AAACCCAAGGAACTCG	0.57432126	0.62077616	-1.1093794	1.05452182
## DYE2111:AAACCCAAGTCATAGA	-0.09057276	0.88887015	-0.1959927	-0.15439309
## DYE2111:AAACCCATCAAGCCTA	0.05352617	0.28112559	0.8263444	0.02845094
## DYE2111:AAACCCATCATGCCGG	0.21949986	0.01755898	0.6662496	0.22143967
## DYE2111:AAACCCATCGGCTTCT	-0.01351606	0.40531804	0.7357822	-0.27291961
##	PC_30	RNA_snn_res.0	seurat_clusters	
## DYE2111:AAACCCAAGAGGTATT	-0.3478537	1	11	
## DYE2111:AAACCCAAGGAACTCG	-2.6899527	1	11	
## DYE2111:AAACCCAAGTCATAGA	-1.9089621	1	4	
## DYE2111:AAACCCATCAAGCCTA	-0.7096458	1	4	
## DYE2111:AAACCCATCATGCCGG	-2.4163980	1	7	
## DYE2111:AAACCCATCGGCTTCT	-2.1976523	1	11	
##	RNA_snn_res.0.1	RNA_snn_res.0.2	RNA_snn_res.0.3	
## DYE2111:AAACCCAAGAGGTATT	2	3	2	
## DYE2111:AAACCCAAGGAACTCG	2	5	5	
## DYE2111:AAACCCAAGTCATAGA	2	3	2	

## DYE2111:AAACCCATCAAGCCTA	2	3	2
## DYE2111:AAACCCATCATGCCGG	2	3	2
## DYE2111:AAACCCATCGGCTTCT	2	5	5
##	RNA_snn_res.0.4	RNA_snn_res.0.5	RNA_snn_res.0.6
## DYE2111:AAACCCAAGAGGTATT	2	2	6
## DYE2111:AAACCCAAGGAACTCG	2	6	6
## DYE2111:AAACCCAAGTCATAGA	2	2	6
## DYE2111:AAACCCATCAAGCCTA	2	2	8
## DYE2111:AAACCCATCATGCCGG	2	2	6
## DYE2111:AAACCCATCGGCTTCT	2	6	7
##	RNA_snn_res.0.7	RNA_snn_res.0.8	RNA_snn_res.0.9
## DYE2111:AAACCCAAGAGGTATT	5	11	3
## DYE2111:AAACCCAAGGAACTCG	5	11	3
## DYE2111:AAACCCAAGTCATAGA	5	4	3
## DYE2111:AAACCCATCAAGCCTA	8	4	6
## DYE2111:AAACCCATCATGCCGG	5	7	3
## DYE2111:AAACCCATCGGCTTCT	6	11	7
##	RNA_snn_res.1	RNA_snn_res.2	RNA_snn_res.3
## DYE2111:AAACCCAAGAGGTATT	3	9	18
## DYE2111:AAACCCAAGGAACTCG	3	9	18
## DYE2111:AAACCCAAGTCATAGA	3	19	22
## DYE2111:AAACCCATCAAGCCTA	7	2	9
## DYE2111:AAACCCATCATGCCGG	3	9	18
## DYE2111:AAACCCATCGGCTTCT	5	9	18
##	RNA_snn_res.4	RNA_snn_res.5	RNA_snn_res.6
## DYE2111:AAACCCAAGAGGTATT	36	36	45
## DYE2111:AAACCCAAGGAACTCG	36	36	45
## DYE2111:AAACCCAAGTCATAGA	13	35	34
## DYE2111:AAACCCATCAAGCCTA	19	44	47
## DYE2111:AAACCCATCATGCCGG	13	35	34
## DYE2111:AAACCCATCGGCTTCT	36	36	45
##	RNA_snn_res.7	RNA_snn_res.8	RNA_snn_res.9
## DYE2111:AAACCCAAGAGGTATT	46	53	48
## DYE2111:AAACCCAAGGAACTCG	46	78	7
## DYE2111:AAACCCAAGTCATAGA	34	35	30
## DYE2111:AAACCCATCAAGCCTA	50	43	78
## DYE2111:AAACCCATCATGCCGG	34	35	30
## DYE2111:AAACCCATCGGCTTCT	46	78	7
##	RNA_snn_res.10	orig.ident.1	nCount_RNA.1
## DYE2111:AAACCCAAGAGGTATT	64	DYE2111	34571
## DYE2111:AAACCCAAGGAACTCG	59	DYE2111	49746
## DYE2111:AAACCCAAGTCATAGA	47	DYE2111	23386
## DYE2111:AAACCCATCAAGCCTA	77	DYE2111	26406
## DYE2111:AAACCCATCATGCCGG	36	DYE2111	25219
## DYE2111:AAACCCATCGGCTTCT	59	DYE2111	50953
##	nFeature_RNA.1	percent.mito.1	log10GenesPerUMI.1
## DYE2111:AAACCCAAGAGGTATT	6588	2.895	0.8413739
## DYE2111:AAACCCAAGGAACTCG	7951	3.052	0.8304498
## DYE2111:AAACCCAAGTCATAGA	5694	2.972	0.8595686
## DYE2111:AAACCCATCAAGCCTA	5455	2.458	0.8451031
## DYE2111:AAACCCATCATGCCGG	5011	2.661	0.8405618

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## DYE2111:AAACCCATCGGCTTCT      8142      2.237      0.8308031
##                               ID.1      SAMPLE.1
## DYE2111:AAACCCAAGAGGTATT DYE2111 3149501_DYE2111
## DYE2111:AAACCCAAGGAACTCG DYE2111 3149501_DYE2111
## DYE2111:AAACCCAAGTCATAGA DYE2111 3149501_DYE2111
## DYE2111:AAACCCATCAAGCCTA DYE2111 3149501_DYE2111
## DYE2111:AAACCCATCATGCCGG DYE2111 3149501_DYE2111
## DYE2111:AAACCCATCGGCTTCT DYE2111 3149501_DYE2111
##
## DYE2111:AAACCCAAGAGGTATT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
1/3149501/
## DYE2111:AAACCCAAGGAACTCG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
1/3149501/
## DYE2111:AAACCCAAGTCATAGA /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
1/3149501/
## DYE2111:AAACCCATCAAGCCTA /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
1/3149501/
## DYE2111:AAACCCATCATGCCGG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
1/3149501/
## DYE2111:AAACCCATCGGCTTCT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
1/3149501/
##                               SJUID.1      SJID.1 cancer_type.1
## DYE2111:AAACCCAAGAGGTATT SJHBBBQCTFK SJEWS056156 Ewing_sarcoma
## DYE2111:AAACCCAAGGAACTCG SJHBBBQCTFK SJEWS056156 Ewing_sarcoma
## DYE2111:AAACCCAAGTCATAGA SJHBBBQCTFK SJEWS056156 Ewing_sarcoma
## DYE2111:AAACCCATCAAGCCTA SJHBBBQCTFK SJEWS056156 Ewing_sarcoma
## DYE2111:AAACCCATCATGCCGG SJHBBBQCTFK SJEWS056156 Ewing_sarcoma
## DYE2111:AAACCCATCGGCTTCT SJHBBBQCTFK SJEWS056156 Ewing_sarcoma
##                               Subject_Disease_Code.1 Subject_Disease_Subtype_Code.1
## DYE2111:AAACCCAAGAGGTATT      EWS      EWS
## DYE2111:AAACCCAAGGAACTCG      EWS      EWS
## DYE2111:AAACCCAAGTCATAGA      EWS      EWS
## DYE2111:AAACCCATCAAGCCTA      EWS      EWS
## DYE2111:AAACCCATCATGCCGG      EWS      EWS
## DYE2111:AAACCCATCGGCTTCT      EWS      EWS
##                               tissue_type.1 assay_technology_updated.1
## DYE2111:AAACCCAAGAGGTATT      CCLF_cells      10x_scRNAseq
## DYE2111:AAACCCAAGGAACTCG      CCLF_cells      10x_scRNAseq
## DYE2111:AAACCCAAGTCATAGA      CCLF_cells      10x_scRNAseq
## DYE2111:AAACCCATCAAGCCTA      CCLF_cells      10x_scRNAseq
## DYE2111:AAACCCATCATGCCGG      CCLF_cells      10x_scRNAseq
## DYE2111:AAACCCATCGGCTTCT      CCLF_cells      10x_scRNAseq
##                               assay_technology.1 historic_data.1 RPMI.1
## DYE2111:AAACCCAAGAGGTATT      10x scRNAseq      no      no
## DYE2111:AAACCCAAGGAACTCG      10x scRNAseq      no      no
## DYE2111:AAACCCAAGTCATAGA      10x scRNAseq      no      no
## DYE2111:AAACCCATCAAGCCTA      10x scRNAseq      no      no
## DYE2111:AAACCCATCATGCCGG      10x scRNAseq      no      no
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##                               genome_reference_updated_code.1
## DYE2111:AAACCCAAGAGGTATT      human-genome

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## DYE2111:AAACCCAAGGAACTCG	human-genome	
## DYE2111:AAACCCAAGTCATAGA	human-genome	
## DYE2111:AAACCCATCAAGCCTA	human-genome	
## DYE2111:AAACCCATCATGCCGG	human-genome	
## DYE2111:AAACCCATCGGCTTCT	human-genome	
##	genome_reference_updated.1	genome_reference.1
## DYE2111:AAACCCAAGAGGTATT	human-genome	dual-genome
## DYE2111:AAACCCAAGGAACTCG	human-genome	dual-genome
## DYE2111:AAACCCAAGTCATAGA	human-genome	dual-genome
## DYE2111:AAACCCATCAAGCCTA	human-genome	dual-genome
## DYE2111:AAACCCATCATGCCGG	human-genome	dual-genome
## DYE2111:AAACCCATCGGCTTCT	human-genome	dual-genome
##		
## DYE2111:AAACCCAAGAGGTATT	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina	
1/		
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## DYE2111:AAACCCAAGTCATAGA	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina	
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## DYE2111:AAACCCATCAAGCCTA	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina	
1/		
## DYE2111:AAACCCATCATGCCGG	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina	
1/		
## DYE2111:AAACCCATCGGCTTCT	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina	
1/		
##	DYExxxx.1	DYE_Melody_unsure.1 DYExxxx_corrected.1
## DYE2111:AAACCCAAGAGGTATT	DYE2111	<NA> DYE2111
## DYE2111:AAACCCAAGGAACTCG	DYE2111	<NA> DYE2111
## DYE2111:AAACCCAAGTCATAGA	DYE2111	<NA> DYE2111
## DYE2111:AAACCCATCAAGCCTA	DYE2111	<NA> DYE2111
## DYE2111:AAACCCATCATGCCGG	DYE2111	<NA> DYE2111
## DYE2111:AAACCCATCGGCTTCT	DYE2111	<NA> DYE2111
##	SRM_updated.1	SRM.1 SRM_Sample_ID_updated.1
## DYE2111:AAACCCAAGAGGTATT	829202	829202 3149501
## DYE2111:AAACCCAAGGAACTCG	829202	829202 3149501
## DYE2111:AAACCCAAGTCATAGA	829202	829202 3149501
## DYE2111:AAACCCATCAAGCCTA	829202	829202 3149501
## DYE2111:AAACCCATCATGCCGG	829202	829202 3149501
## DYE2111:AAACCCATCGGCTTCT	829202	829202 3149501
##	SRM_Sample_ID.1	Experimental_Sample.1 sample_3D.1
## DYE2111:AAACCCAAGAGGTATT	3149501	SJ1044 No
## DYE2111:AAACCCAAGGAACTCG	3149501	SJ1044 No
## DYE2111:AAACCCAAGTCATAGA	3149501	SJ1044 No
## DYE2111:AAACCCATCAAGCCTA	3149501	SJ1044 No
## DYE2111:AAACCCATCATGCCGG	3149501	SJ1044 No
## DYE2111:AAACCCATCGGCTTCT	3149501	SJ1044 No
##	sample_2D.NOT.NEEDED.2025.04.10.1	Tissue_Bank_ID.1
## DYE2111:AAACCCAAGAGGTATT	No	TB-19-21394
## DYE2111:AAACCCAAGGAACTCG	No	TB-19-21394
## DYE2111:AAACCCAAGTCATAGA	No	TB-19-21394
## DYE2111:AAACCCATCAAGCCTA	No	TB-19-21394

## DYE2111:AAACCCATCATGCCGG	No	TB-19-21394
## DYE2111:AAACCCATCGGCTTCT	No	TB-19-21394
##	Xenograft_CompBio_ID.1	CSTN_MAST_ID.1
## DYE2111:AAACCCAAGAGGTATT	SJEWS056156_X1	MAST601E
## DYE2111:AAACCCAAGGAACTCG	SJEWS056156_X1	MAST601E
## DYE2111:AAACCCAAGTCATAGA	SJEWS056156_X1	MAST601E
## DYE2111:AAACCCATCAAGCCTA	SJEWS056156_X1	MAST601E
## DYE2111:AAACCCATCATGCCGG	SJEWS056156_X1	MAST601E
## DYE2111:AAACCCATCGGCTTCT	SJEWS056156_X1	MAST601E
##	Date.1	SRM_Submission_Information.1
## DYE2111:AAACCCAAGAGGTATT	2024-06-14T00:00:00Z	<NA>
## DYE2111:AAACCCAAGGAACTCG	2024-06-14T00:00:00Z	<NA>
## DYE2111:AAACCCAAGTCATAGA	2024-06-14T00:00:00Z	<NA>
## DYE2111:AAACCCATCAAGCCTA	2024-06-14T00:00:00Z	<NA>
## DYE2111:AAACCCATCATGCCGG	2024-06-14T00:00:00Z	<NA>
## DYE2111:AAACCCATCGGCTTCT	2024-06-14T00:00:00Z	<NA>
##	Compbioname_for_publications.1	
## DYE2111:AAACCCAAGAGGTATT	SJEWS056156_C3	
## DYE2111:AAACCCAAGGAACTCG	SJEWS056156_C3	
## DYE2111:AAACCCAAGTCATAGA	SJEWS056156_C3	
## DYE2111:AAACCCATCAAGCCTA	SJEWS056156_C3	
## DYE2111:AAACCCATCATGCCGG	SJEWS056156_C3	
## DYE2111:AAACCCATCGGCTTCT	SJEWS056156_C3	
##	Comp_Bio_Sample_Name.1	Comp_Bio_Subject_ID.1
## DYE2111:AAACCCAAGAGGTATT	SJEWS056156_C3-SJ1044T_2	SJ056156
## DYE2111:AAACCCAAGGAACTCG	SJEWS056156_C3-SJ1044T_2	SJ056156
## DYE2111:AAACCCAAGTCATAGA	SJEWS056156_C3-SJ1044T_2	SJ056156
## DYE2111:AAACCCATCAAGCCTA	SJEWS056156_C3-SJ1044T_2	SJ056156
## DYE2111:AAACCCATCATGCCGG	SJEWS056156_C3-SJ1044T_2	SJ056156
## DYE2111:AAACCCATCGGCTTCT	SJEWS056156_C3-SJ1044T_2	SJ056156
##	Preferred_Sample_ID.1	Sample_Type_Code.1
## DYE2111:AAACCCAAGAGGTATT	SJ1044T_2	C
## DYE2111:AAACCCAAGGAACTCG	SJ1044T_2	C
## DYE2111:AAACCCAAGTCATAGA	SJ1044T_2	C
## DYE2111:AAACCCATCAAGCCTA	SJ1044T_2	C
## DYE2111:AAACCCATCATGCCGG	SJ1044T_2	C
## DYE2111:AAACCCATCGGCTTCT	SJ1044T_2	C
##	Prior_Comp_Bio_ID.1	PATH.1 NOTES.1 match_SJID_RPMI.1
## DYE2111:AAACCCAAGAGGTATT	SJEWS056156	<NA> NA SJEWS056156_no
## DYE2111:AAACCCAAGGAACTCG	SJEWS056156	<NA> NA SJEWS056156_no
## DYE2111:AAACCCAAGTCATAGA	SJEWS056156	<NA> NA SJEWS056156_no
## DYE2111:AAACCCATCAAGCCTA	SJEWS056156	<NA> NA SJEWS056156_no
## DYE2111:AAACCCATCATGCCGG	SJEWS056156	<NA> NA SJEWS056156_no
## DYE2111:AAACCCATCGGCTTCT	SJEWS056156	<NA> NA SJEWS056156_no
##	ID_with_sample_swap.1	SAMPLE_with_sample_swap.1
## DYE2111:AAACCCAAGAGGTATT	DYE2111	3149501_DYE2111
## DYE2111:AAACCCAAGGAACTCG	DYE2111	3149501_DYE2111
## DYE2111:AAACCCAAGTCATAGA	DYE2111	3149501_DYE2111
## DYE2111:AAACCCATCAAGCCTA	DYE2111	3149501_DYE2111
## DYE2111:AAACCCATCATGCCGG	DYE2111	3149501_DYE2111
## DYE2111:AAACCCATCGGCTTCT	DYE2111	3149501_DYE2111

```

##
## DYE2111:AAACCCAAGAGGTATT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
1/3149501/
## DYE2111:AAACCCAAGGAACTCG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
1/3149501/
## DYE2111:AAACCCAAGTCATAGA /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
1/3149501/
## DYE2111:AAACCCATCAAGCCTA /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
1/3149501/
## DYE2111:AAACCCATCATGCCGG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
1/3149501/
## DYE2111:AAACCCATCGGCTTCT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
1/3149501/
##
## S.Score.1 G2M.Score.1 Phase.1 UMAP_1.x UMAP_2.x
## DYE2111:AAACCCAAGAGGTATT 0.055396504 0.5217267 G2M 0.7912828 6.248889
## DYE2111:AAACCCAAGGAACTCG 0.029724122 0.4497271 G2M 1.9758112 7.561102
## DYE2111:AAACCCAAGTCATAGA 0.003884657 0.1267804 G2M 1.9600578 4.757619
## DYE2111:AAACCCATCAAGCCTA 0.194223495 -0.2824893 S 4.2224048 5.308984
## DYE2111:AAACCCATCATGCCGG 0.005253221 0.4780977 G2M 1.5425332 5.045206
## DYE2111:AAACCCATCGGCTTCT -0.060120228 0.3925691 G2M 2.1266411 8.001167
##
## PC_1.x PC_2.x PC_3.x PC_4.x PC_5.x
## DYE2111:AAACCCAAGAGGTATT 10.253152 -22.68979 -0.6323919 -0.9587824 8.769602
## DYE2111:AAACCCAAGGAACTCG 8.485662 -24.39137 0.8013676 -1.4042601 10.082944
## DYE2111:AAACCCAAGTCATAGA 10.951001 -20.07229 -1.7588574 0.2910809 7.229904
## DYE2111:AAACCCATCAAGCCTA 11.597054 -18.91220 -0.3046808 1.4613042 4.900138
## DYE2111:AAACCCATCATGCCGG 13.183820 -23.07905 -3.7541129 -0.3845318 10.246315
## DYE2111:AAACCCATCGGCTTCT 8.375762 -21.93434 2.5777548 -1.4043628 7.400935
##
## PC_6.x PC_7.x PC_8.x PC_9.x
## DYE2111:AAACCCAAGAGGTATT -2.7712517 -9.366240 -1.4974979 2.72132666
## DYE2111:AAACCCAAGGAACTCG -0.3636017 -12.209703 -4.7558793 1.74341078
## DYE2111:AAACCCAAGTCATAGA 3.1424492 -7.601283 -0.1429734 2.26775331
## DYE2111:AAACCCATCAAGCCTA 8.6664426 -8.547095 -3.4713420 0.05574464
## DYE2111:AAACCCATCATGCCGG 0.3128649 -7.595591 1.7437701 1.86239495
## DYE2111:AAACCCATCGGCTTCT -0.9476664 -11.345271 -3.7922004 0.40670382
##
## PC_10.x PC_11.x PC_12.x PC_13.x
## DYE2111:AAACCCAAGAGGTATT 1.23112598 1.582408 -0.46662568 -2.789662
## DYE2111:AAACCCAAGGAACTCG 3.70481977 3.357483 -0.22436947 -3.751007
## DYE2111:AAACCCAAGTCATAGA 0.08998602 2.052672 -0.52955431 -3.742995
## DYE2111:AAACCCATCAAGCCTA -3.97244738 3.535814 0.08966243 -1.188321
## DYE2111:AAACCCATCATGCCGG 1.07962243 -1.074452 -0.84847652 -2.609866
## DYE2111:AAACCCATCGGCTTCT 3.38565479 3.496356 -1.30506804 -3.548699
##
## PC_14.x PC_15.x PC_16.x PC_17.x
## DYE2111:AAACCCAAGAGGTATT 0.126148021 -0.84918671 1.736273165 -1.7131367
## DYE2111:AAACCCAAGGAACTCG 0.059867572 0.38250345 -0.004923742 -0.6104021
## DYE2111:AAACCCAAGTCATAGA 0.324806608 0.97984809 0.379255062 -0.4690637
## DYE2111:AAACCCATCAAGCCTA -0.667426810 -0.50624282 0.038240148 -0.8174554
## DYE2111:AAACCCATCATGCCGG -0.006686692 -0.01481017 -0.599154869 -0.4504755
## DYE2111:AAACCCATCGGCTTCT 0.828032768 0.48924571 1.251134631 -0.4071848
##
## PC_18.x PC_19.x PC_20.x PC_21.x
## DYE2111:AAACCCAAGAGGTATT 0.3221185 -0.98278867 1.7082152 -1.4219440
## DYE2111:AAACCCAAGGAACTCG -1.7462241 0.70255332 1.1868776 -1.8061401

```


## DYE2111:AAACCCAAGTCATAGA	0.1839820	0.05369649	1.1130911	0.2894633
## DYE2111:AAACCCATCAAGCCTA	0.8179425	0.31679649	0.5496390	-0.9057892
## DYE2111:AAACCCATCATGCCGG	-1.2985162	-1.58671698	0.3389435	1.0787569
## DYE2111:AAACCCATCGGCTTCT	-1.4768153	-0.73618128	0.3753112	-0.6263584
##	PC_22.x	PC_23.x	PC_24.x	PC_25.x
## DYE2111:AAACCCAAGAGGTATT	0.04758052	-0.75999468	-0.6758716	-1.03014705
## DYE2111:AAACCCAAGGAACTCG	-2.06483099	-0.04886289	-1.6141406	-0.34460262
## DYE2111:AAACCCAAGTCATAGA	-0.92594862	-0.67978153	-0.1869200	0.72142182
## DYE2111:AAACCCATCAAGCCTA	-0.18961110	0.69286040	-1.2093640	0.46370255
## DYE2111:AAACCCATCATGCCGG	0.07483081	-0.72005199	0.2030740	-0.93556990
## DYE2111:AAACCCATCGGCTTCT	0.05809369	0.46735142	-0.2989385	-0.08005591
##	PC_26.x	PC_27.x	PC_28.x	PC_29.x
## DYE2111:AAACCCAAGAGGTATT	0.68494819	-1.21100072	-0.6489976	-1.04018690
## DYE2111:AAACCCAAGGAACTCG	0.57432126	0.62077616	-1.1093794	1.05452182
## DYE2111:AAACCCAAGTCATAGA	-0.09057276	0.88887015	-0.1959927	-0.15439309
## DYE2111:AAACCCATCAAGCCTA	0.05352617	0.28112559	0.8263444	0.02845094
## DYE2111:AAACCCATCATGCCGG	0.21949986	0.01755898	0.6662496	0.22143967
## DYE2111:AAACCCATCGGCTTCT	-0.01351606	0.40531804	0.7357822	-0.27291961
##	PC_30.x	RNA_snn_res.0.10	seurat_clusters.1	
## DYE2111:AAACCCAAGAGGTATT	-0.3478537	1	64	
## DYE2111:AAACCCAAGGAACTCG	-2.6899527	1	59	
## DYE2111:AAACCCAAGTCATAGA	-1.9089621	1	47	
## DYE2111:AAACCCATCAAGCCTA	-0.7096458	1	77	
## DYE2111:AAACCCATCATGCCGG	-2.4163980	1	36	
## DYE2111:AAACCCATCGGCTTCT	-2.1976523	1	59	
##	RNA_snn_res.0.1.1	RNA_snn_res.0.2.1	RNA_snn_res.0.3.1	
## DYE2111:AAACCCAAGAGGTATT	2	3	2	
## DYE2111:AAACCCAAGGAACTCG	2	5	5	
## DYE2111:AAACCCAAGTCATAGA	2	3	2	
## DYE2111:AAACCCATCAAGCCTA	2	3	2	
## DYE2111:AAACCCATCATGCCGG	2	3	2	
## DYE2111:AAACCCATCGGCTTCT	2	5	5	
##	RNA_snn_res.0.4.1	RNA_snn_res.0.5.1	RNA_snn_res.0.6.1	
## DYE2111:AAACCCAAGAGGTATT	2	2	6	
## DYE2111:AAACCCAAGGAACTCG	2	6	6	
## DYE2111:AAACCCAAGTCATAGA	2	2	6	
## DYE2111:AAACCCATCAAGCCTA	2	2	8	
## DYE2111:AAACCCATCATGCCGG	2	2	6	
## DYE2111:AAACCCATCGGCTTCT	2	6	7	
##	RNA_snn_res.0.7.1	RNA_snn_res.0.8.1	RNA_snn_res.0.9.1	
## DYE2111:AAACCCAAGAGGTATT	5	5	3	
## DYE2111:AAACCCAAGGAACTCG	5	5	3	
## DYE2111:AAACCCAAGTCATAGA	5	5	3	
## DYE2111:AAACCCATCAAGCCTA	8	8	6	
## DYE2111:AAACCCATCATGCCGG	5	5	3	
## DYE2111:AAACCCATCGGCTTCT	6	4	7	
##	RNA_snn_res.1.1	RNA_snn_res.2.1	RNA_snn_res.3.1	
## DYE2111:AAACCCAAGAGGTATT	3	9	18	
## DYE2111:AAACCCAAGGAACTCG	3	9	18	
## DYE2111:AAACCCAAGTCATAGA	3	19	22	
## DYE2111:AAACCCATCAAGCCTA	7	2	9	

## DYE2111:AAACCCATCATGCCGG	3	9	18		
## DYE2111:AAACCCATCGGCTTCT	5	9	18		
##	RNA_snn_res.4.1	RNA_snn_res.5.1	RNA_snn_res.6.1		
## DYE2111:AAACCCAAGAGGTATT	36	36	45		
## DYE2111:AAACCCAAGGAACTCG	36	36	45		
## DYE2111:AAACCCAAGTCATAGA	13	35	34		
## DYE2111:AAACCCATCAAGCCTA	19	44	47		
## DYE2111:AAACCCATCATGCCGG	13	35	34		
## DYE2111:AAACCCATCGGCTTCT	36	36	45		
##	RNA_snn_res.7.1	RNA_snn_res.8.1	RNA_snn_res.9.1		
## DYE2111:AAACCCAAGAGGTATT	46	53	48		
## DYE2111:AAACCCAAGGAACTCG	46	78	7		
## DYE2111:AAACCCAAGTCATAGA	34	35	30		
## DYE2111:AAACCCATCAAGCCTA	50	43	78		
## DYE2111:AAACCCATCATGCCGG	34	35	30		
## DYE2111:AAACCCATCGGCTTCT	46	78	7		
##	RNA_snn_res.10.1	UMAP_1.y	UMAP_2.y	PC_1.y	PC_2.y
## DYE2111:AAACCCAAGAGGTATT	64	-5.309351	4.834933	12.61036	16.02497
## DYE2111:AAACCCAAGGAACTCG	59	-4.689333	5.448497	11.28931	19.11555
## DYE2111:AAACCCAAGTCATAGA	47	-4.454633	3.344939	13.09062	14.06508
## DYE2111:AAACCCATCAAGCCTA	77	-2.375616	4.372145	13.78687	13.24313
## DYE2111:AAACCCATCATGCCGG	36	-5.149113	2.321172	15.75144	15.07727
## DYE2111:AAACCCATCGGCTTCT	59	-4.577039	6.124684	10.42014	17.53970
##	PC_3.y	PC_4.y	PC_5.y	PC_6.y	PC_7.y
## DYE2111:AAACCCAAGAGGTATT	-8.024592	8.499704	5.3538917	-7.054619	2.871856885
## DYE2111:AAACCCAAGGAACTCG	-8.218600	9.776419	3.3505747	-11.916284	3.459336053
## DYE2111:AAACCCAAGTCATAGA	-7.872348	8.893445	-0.7526152	-5.343806	1.507516793
## DYE2111:AAACCCATCAAGCCTA	-5.743224	8.832146	-6.8231766	-7.466742	1.923006392
## DYE2111:AAACCCATCATGCCGG	-11.371866	11.359989	2.7862604	-3.559772	0.006959211
## DYE2111:AAACCCATCGGCTTCT	-5.047124	6.712116	3.5824271	-11.631276	2.167243496
##	PC_8.y	PC_9.y	PC_10.y	PC_11.y	PC_12.y
## DYE2111:AAACCCAAGAGGTATT	-1.2471564	0.4498646	-2.0314513	2.2465921	-
0.54016960					
## DYE2111:AAACCCAAGGAACTCG	-2.5812647	0.6883219	-2.0165538	2.8665437	0.18502043
## DYE2111:AAACCCAAGTCATAGA	-0.6182391	1.4162367	-3.3086952	3.1722259	1.92587871
## DYE2111:AAACCCATCAAGCCTA	3.2350795	4.0119031	-0.8201189	0.7741291	-
0.62852155					
## DYE2111:AAACCCATCATGCCGG	-0.6636783	-1.1413408	-1.1470311	2.3701901	0.07340171
## DYE2111:AAACCCATCGGCTTCT	-2.2050135	0.6768091	-2.4841288	2.4817824	0.83562270
##	PC_13.y	PC_14.y	PC_15.y	PC_16.y	
## DYE2111:AAACCCAAGAGGTATT	0.8645031	-1.7908525	-1.36973854	-0.28247081	
## DYE2111:AAACCCAAGGAACTCG	1.5294162	0.4244165	-0.63493385	-0.05967588	
## DYE2111:AAACCCAAGTCATAGA	0.9853435	-0.5766405	-0.06959485	0.33712090	
## DYE2111:AAACCCATCAAGCCTA	0.2915780	-0.2702213	-0.82280802	1.16607726	
## DYE2111:AAACCCATCATGCCGG	1.4599301	1.0154368	-1.56496166	-1.09157039	
## DYE2111:AAACCCATCGGCTTCT	0.9890064	-0.5921915	-0.02836556	-1.79817606	
##	PC_17.y	PC_18.y	PC_19.y	PC_20.y	
## DYE2111:AAACCCAAGAGGTATT	0.7337291	-1.2152835	0.7585652	0.37836499	
## DYE2111:AAACCCAAGGAACTCG	-1.5949383	0.1071244	2.0516295	-2.15063902	
## DYE2111:AAACCCAAGTCATAGA	0.2377252	-0.7773358	0.2095904	-0.69637966	
## DYE2111:AAACCCATCAAGCCTA	0.4999611	0.1887554	1.4689274	-0.38521171	

```

## DYE2111:AAACCCATCATGCCGG 0.7187847 -0.2171192 -1.8466226 -0.04339832
## DYE2111:AAACCCATCGGCTTCT -0.5026278 -0.3541162 0.8875527 -0.31061422
##
## PC_21.y PC_22.y PC_23.y PC_24.y
## DYE2111:AAACCCAAGAGGTATT 0.20767042 -0.3212317 -1.1301084 -2.20913317
## DYE2111:AAACCCAAGGAAGTCTG -0.07706783 -0.4920501 -0.1376365 -0.13267593
## DYE2111:AAACCCAAGTCATAGA 0.41428929 0.6441728 -0.9224697 -0.11543797
## DYE2111:AAACCCATCAAGCCTA 0.85025770 -0.4906410 0.1572436 0.45990651
## DYE2111:AAACCCATCATGCCGG -0.21839859 -0.3079748 -0.3562500 0.06077707
## DYE2111:AAACCCATCGGCTTCT 0.20565321 -0.2796623 0.3611114 0.47606127
##
## PC_25.y PC_26.y PC_27.y PC_28.y
## DYE2111:AAACCCAAGAGGTATT 0.3429889 0.0155035 -0.6566335 0.17339325
## DYE2111:AAACCCAAGGAAGTCTG -0.6611147 -0.1256383 -2.1257335 1.53594388
## DYE2111:AAACCCAAGTCATAGA -1.1701945 -1.0910408 -0.5882027 1.03552281
## DYE2111:AAACCCATCAAGCCTA 1.5394916 0.3965679 -1.0582192 0.56391982
## DYE2111:AAACCCATCATGCCGG -0.1324925 0.1345148 -2.4746101 -0.09487223
## DYE2111:AAACCCATCGGCTTCT 0.3874799 -0.4065605 -2.3404917 -0.49024907
##
## PC_29.y PC_30.y
## DYE2111:AAACCCAAGAGGTATT -0.06346352 -1.7324476
## DYE2111:AAACCCAAGGAAGTCTG -1.68529714 0.4431799
## DYE2111:AAACCCAAGTCATAGA -1.81554426 -0.2685423
## DYE2111:AAACCCATCAAGCCTA -1.38853362 -0.8529847
## DYE2111:AAACCCATCATGCCGG -0.56067207 -0.7384978
## DYE2111:AAACCCATCGGCTTCT 0.17673850 -1.3569457

```

```

saveRDS(seurat_obj, file = paste0(results_dir, "/", glue::glue("seurat_obj_integrated_{inte

```

10 Session Info

```
## R version 4.4.0 (2024-04-24)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 22.04.4 LTS
##
## Matrix products: default
## BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p0.3.20.so; LAPACK vers
##
## locale:
## [1] C
##
## time zone: Etc/UTC
## tzcode source: system (glibc)
##
## attached base packages:
## [1] grid      stats      graphics  grDevices  utils      datasets  methods
## [8] base
##
## other attached packages:
## [1] knitr_1.49          reshape2_1.4.4      SeuratWrappers_0.2.0
## [4] RcppPlanc_1.0.0     rliger_2.1.0.9004   harmony_1.2.3
## [7] Rcpp_1.0.14         future_1.34.0       RColorBrewer_1.1-3
## [10] ggthemes_5.1.0      patchwork_1.3.0     scooter_0.0.0.9004
## [13] SeuratObject_5.0.2  Seurat_4.4.0        lubridate_1.9.4
## [16] forcats_1.0.0       stringr_1.5.1       dplyr_1.1.4
## [19] purrr_1.0.2         readr_2.1.5         tidyr_1.3.1
## [22] tibble_3.2.1        ggplot2_3.5.1       tidyverse_2.0.0
## [25] yaml_2.3.10
##
## loaded via a namespace (and not attached):
## [1] jsonlite_1.8.9      magrittr_2.0.3      ggbeeswarm_0.7.2
## [4] spatstat.utils_3.1-2 farver_2.1.2         rmarkdown_2.29
## [7] ragg_1.3.3          vctrs_0.6.5         ROCR_1.0-11
## [10] spatstat.explore_3.3-4 htmltools_0.5.8.1    sass_0.4.9
## [13] sctransform_0.4.1    parallelly_1.42.0    bslib_0.9.0
## [16] KernSmooth_2.23-26   htmlwidgets_1.6.4    ica_1.0-3
## [19] plyr_1.8.9          cachem_1.1.0         plotly_4.10.4
## [22] zoo_1.8-12          igraph_2.1.4         mime_0.12
## [25] lifecycle_1.0.4     pkgconfig_2.0.3      rsvd_1.0.5
## [28] Matrix_1.7-2        R6_2.5.1             fastmap_1.2.0
## [31] fitdistrplus_1.2-2   shiny_1.10.0         digest_0.6.37
## [34] colorspace_2.1-1     S4Vectors_0.44.0     tensor_1.5
## [37] irlba_2.3.5.1        textshaping_1.0.0    labeling_0.4.3
## [40] progressr_0.15.1     spatstat.sparse_3.1-0 timechange_0.3.0
## [43] httr_1.4.7          polyclip_1.10-7      abind_1.4-8
## [46] compiler_4.4.0       remotes_2.5.0        bit64_4.6.0-1
## [49] withr_3.0.2          MASS_7.3-64          ggsci_3.2.0
## [52] tools_4.4.0          vipor_0.4.7          lmtest_0.9-40
## [55] beeswarm_0.4.0       httpuv_1.6.15        future.apply_1.11.3
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## [58] goftest_1.2-3	glue_1.8.0	nlme_3.1-167
## [61] promises_1.3.2	Rtsne_0.17	cluster_2.1.8
## [64] generics_0.1.3	gtable_0.3.6	spatstat.data_3.1-4
## [67] tzdb_0.4.0	data.table_1.16.4	hms_1.1.3
## [70] sp_2.2-0	BiocGenerics_0.52.0	spatstat.geom_3.3-5
## [73] RcppAnnoy_0.0.22	ggrepel_0.9.6	RANN_2.6.2
## [76] pillar_1.10.1	spam_2.11-1	vroom_1.6.5
## [79] later_1.4.1	splines_4.4.0	lattice_0.22-6
## [82] survival_3.8-3	bit_4.5.0.1	deldir_2.0-4
## [85] tidyselect_1.2.1	miniUI_0.1.1.1	pbapply_1.7-2
## [88] gridExtra_2.3	scattermore_1.2	RhpcBLASctl_0.23-42
## [91] stats4_4.4.0	xfun_0.50	matrixStats_1.5.0
## [94] stringi_1.8.4	lazyeval_0.2.2	evaluate_1.0.3
## [97] codetools_0.2-20	BiocManager_1.30.25	cli_3.6.3
## [100] uwot_0.2.2	xtable_1.8-4	reticulate_1.40.0
## [103] systemfonts_1.2.1	jquerylib_0.1.4	munsell_0.5.1
## [106] globals_0.16.3	spatstat.random_3.3-2	png_0.1-8
## [109] ggtrastr_1.0.2	spatstat.univar_3.1-1	parallel_4.4.0
## [112] dotCall64_1.2	listenv_0.9.1	viridisLite_0.4.2
## [115] scales_1.3.0	ggiridges_0.5.6	leiden_0.4.3.1
## [118] crayon_1.5.3	rlang_1.1.5	cowplot_1.1.3